

Appendix A: Failure Modes and Containment

Spectral and Quantum Dysentery

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1 Notation and Standing Hypotheses

Let $\Omega \subset \mathbb{R}^d$ be bounded with sufficiently regular boundary. Denote by

$$A_0 := -\Delta_D + \mu I, \quad \mu > 0,$$

the Dirichlet Schrödinger base operator on $L^2(\Omega)$ with compact resolvent, and let

$$A := A_0 + V, \quad V \in L^\infty(\Omega; \mathbb{R}),$$

be a bounded self-adjoint perturbation. Write $\lambda_1(-\Delta_D)$ for the first Dirichlet eigenvalue and define

$$c_* := \lambda_1(-\Delta_D) + \mu - \|V\|_{L^\infty(\Omega)}.$$

Throughout, $\Delta(A)$ denotes the spectral gap above the ground level.

2 Definitions

Definition 2.1 (Spectral Dysentery). *Let A be a closed, densely defined operator on a Hilbert space $\mathcal{H}$ with ground level $\lambda_0 = \inf \text{Spec}(A)$. We say that A exhibits spectral dysentery if any of the following holds:*

1. **Gap collapse:** $\Delta(A) = 0$ or $\Delta(A)$ is insufficient to control excited-sector leakage;
2. **Loss of isolation:** the protected spectral component is no longer isolated;
3. **Mode leakage:** under the semigroup e^{-tA} , mass/energy escapes the protected sector at a rate incompatible with exponential confinement;
4. **Resolvent blow-up:** there exists a sequence $z_n \rightarrow \lambda_0$ with $\|(A - z_n I)^{-1}\| \rightarrow \infty$.

Informal: the spectrum fails to contain the dynamics; coherent evolution is replaced by leakage and loss of structure.

Definition 2.2 (Quantum Dysentery). *Let $\rho(t)$ be a density operator on $\mathcal{H}$. The system exhibits quantum dysentery if coherence and protected occupation degrade rapidly, characterized by:*

1. **Decoherence:** off-diagonal decay $|\rho_{ij}(t)| \rightarrow 0$ in a relevant basis;
2. **Entropy growth:** $S(\rho(t)) = -\text{Tr}(\rho(t) \log \rho(t))$ increases toward a high-entropy mixed state;
3. **Loss of protection:** depletion of the protected-mode population;
4. **Irreversibility:** drift to a mixed state not recoverable by unitary control.

Informal: phase information and coherent occupation are flushed into noise.

3 Containment via Spectral Barriers

Theorem 3.1 (Anti-Dysentery Barrier). *Assume $c_* > 0$. Then:*

1. $\inf \text{Spec}(A) \geq c_*$;
2. $\|e^{-tA}\| \leq e^{-c_*t}$ for all $t \geq 0$;
3. for every $z \in \mathbb{C}$ with $\Re z < c_*$,

$$\|(A - zI)^{-1}\| \leq \frac{1}{c_* - \Re z}.$$

Consequently, A is protected against spectral dysentery below the barrier c_* .

Proof. For $u \in H_0^1(\Omega)$,

$$\langle Au, u \rangle = \int_{\Omega} |\nabla u|^2 dx + \mu \|u\|_{L^2}^2 + \int_{\Omega} V|u|^2 dx \geq (\lambda_1(-\Delta_D) + \mu - \|V\|_{L^\infty}) \|u\|_{L^2}^2 = c_* \|u\|_{L^2}^2.$$

Thus $A \geq c_*I$ in the quadratic form sense, yielding the spectral bound. The semigroup and resolvent estimates follow from the spectral theorem. $\square$

Corollary 3.2 (Exponential Confinement). *Under the hypotheses of Theorem 3.1, any solution $u(t) = e^{-tA}u_0$ satisfies $\|u(t)\|_{L^2} \leq e^{-c_*t}\|u_0\|_{L^2}$, preventing sustained leakage into unstable modes.*

4 Thermal Suppression and Quantum Stability

Theorem 4.1 (Exponential Suppression of Excited Sector). *Let $A^{(\mu)}$ be as defined with gap $\Delta^{(\mu)}$. Then for $\beta > 0$,*

$$Z(\beta) := \text{Tr}(e^{-\beta A^{(\mu)}}) \leq e^{-\beta \lambda_0^{(\mu)}} \left(1 + Ce^{-\beta \Delta^{(\mu)}}\right).$$

In particular, if $\beta \Delta^{(\mu)} \gg 1$, excited-state occupation is exponentially suppressed, preventing quantum dysentery.

Proof. Decompose spectrally:

$$Z(\beta) = e^{-\beta \lambda_0^{(\mu)}} + \sum_{n \geq 1} e^{-\beta \lambda_n^{(\mu)}} \leq e^{-\beta \lambda_0^{(\mu)}} + e^{-\beta(\lambda_0^{(\mu)} + \Delta^{(\mu)})} \sum_{n \geq 1} a_n.$$

$\square$

5 Discussion (Informal Synthesis)

Remark 5.1 (Oregon–Trail Formulation). *In deliberately informal language: gap collapse, mode leakage, and loss of resolvent control constitute spectral dysentery; rapid decoherence and entropy growth constitute quantum dysentery. Large spectral barriers and gap amplification are the corresponding anti-dysentery protocol.*